PROJECT REVIEW | PROBABILITY & STATISTICS (BAMAT207)

Breast Cancer Classification using Probabilistic Machine Learning

Track 6 - AI, Machine Learning & Probability

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569

tumour samples, 30 features each

98.25%

test accuracy, Logistic Regression

0.9954

ROC-AUC on the held-out test set

Problem statement and objectives

The problem

A fine-needle biopsy produces measurements of cell nuclei. Deciding malignant versus benign from those numbers is a classification problem, and the answer should carry a probability, not just a label.

PROBLEM STATEMENT

Given 30 numeric features from a digitised biopsy image, predict whether the tumour is malignant or benign, and quantify the uncertainty with probability.

1. **Compare two probabilistic models.** Logistic Regression (maximum likelihood) against Gaussian Naive Bayes (Bayes' theorem).
2. **Make the probability concepts visible.** Conditional probability, Gaussian distribution, MLE, log-odds and Bayesian inference, each shown in real output.
3. **Evaluate honestly.** Held-out test set, confusion matrix, precision and recall, ROC-AUC and 5-fold cross-validation.
4. **Interpret and criticise.** Read coefficients as odds ratios and say where the models fail.

Dataset: Wisconsin Diagnostic Breast Cancer

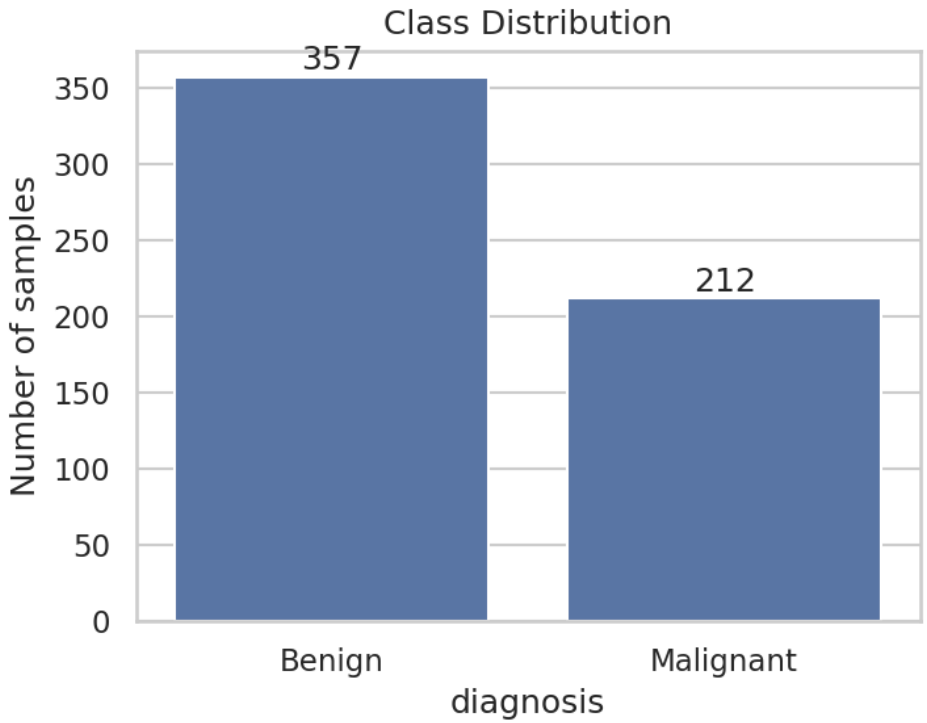
569
tumour samples

30
numeric features

212 / 357
malignant / benign

0
missing values

Each sample has 10 nucleus characteristics (radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, fractal dimension) in 3 forms: mean, standard error and worst.



A model that always says "benign" already scores 62.7%, so accuracy alone is not enough.

Source: Wolberg, Mangasarian, Street & Street (1995), Breast Cancer Wisconsin (Diagnostic), UCI Machine Learning Repository; loaded via scikit-learn.

Probability concepts used in this project

Conditional probability

$$P(A \mid B) = P(A \cap B) / P(B)$$

Precision, recall and specificity are conditional probabilities read straight off the confusion matrix.

Gaussian distribution

$$f(x) = 1 / (\sigma\sqrt{2\pi}) \cdot e^{-(x-\mu)^2 / 2\sigma^2}$$

Class-conditional likelihood of every feature in Gaussian Naive Bayes; checked on the EDA histograms.

Maximum likelihood and log-odds

$$\log L(\beta) = \Sigma [\, y \ln p + (1 - y) \ln(1 - p) \,]$$

Fits Logistic Regression: coefficients are log-odds chosen to make the observed labels most probable.

Bayesian inference

$$P(C \mid x) = P(x \mid C) \cdot P(C) / P(x)$$

Turns the class prior into a posterior probability of malignancy for each patient.

Logistic regression and maximum likelihood

1 Linear score = log-odds

$$z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_{30} x_{30}$$

x = the 30 standardised features

2 Sigmoid turns it into a probability

$$P(\text{benign} \mid x) = 1 / (1 + e^{-z})$$

Always between 0 and 1; a threshold of 0.5 gives the class

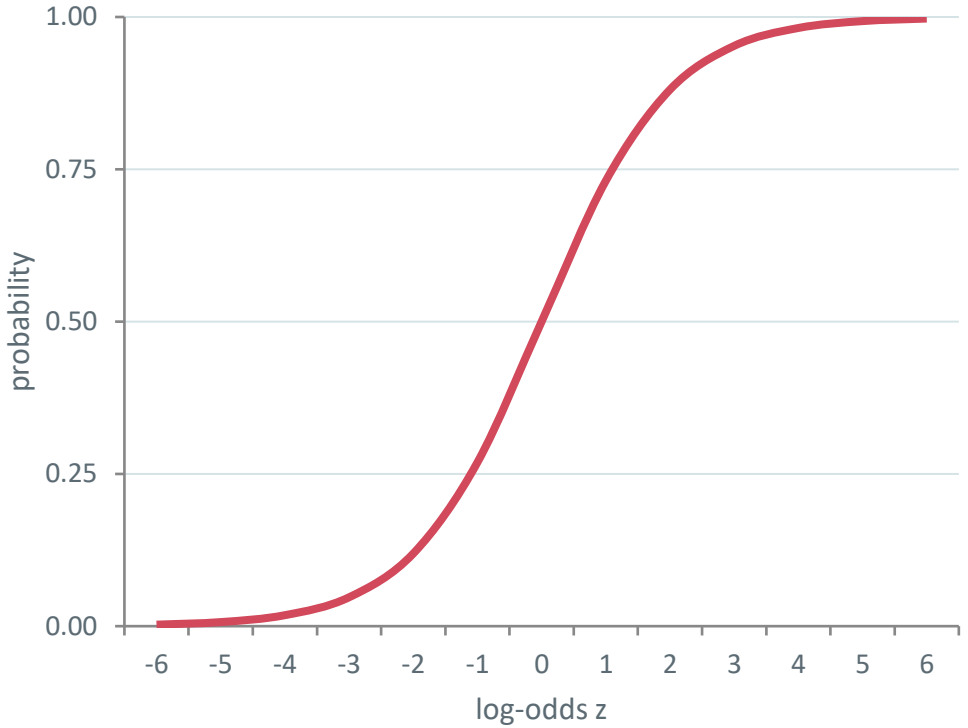
3 Maximum likelihood chooses β

$$\log L(\beta) = \Sigma [\, y \ln p + (1 - y) \ln(1 - p) \,]$$

Maximising log L is the same as minimising log-loss

scikit-learn adds an L2 penalty by default (C = 1.0), so strictly this is penalised maximum likelihood. Fitted intercept β_0 = 0.3022; the solver converged in 19 iterations.

Sigmoid: log-odds z to probability



At z = 0 the probability is exactly 0.5; the further z is from 0, the more confident the prediction.

Gaussian Naive Bayes and Bayesian updating

PRIOR

$$P(\text{malignant}) = 0.3736$$

Class frequency in the training set

×

LIKELIHOOD

$$\prod P(\text{feature} \mid \text{class})$$

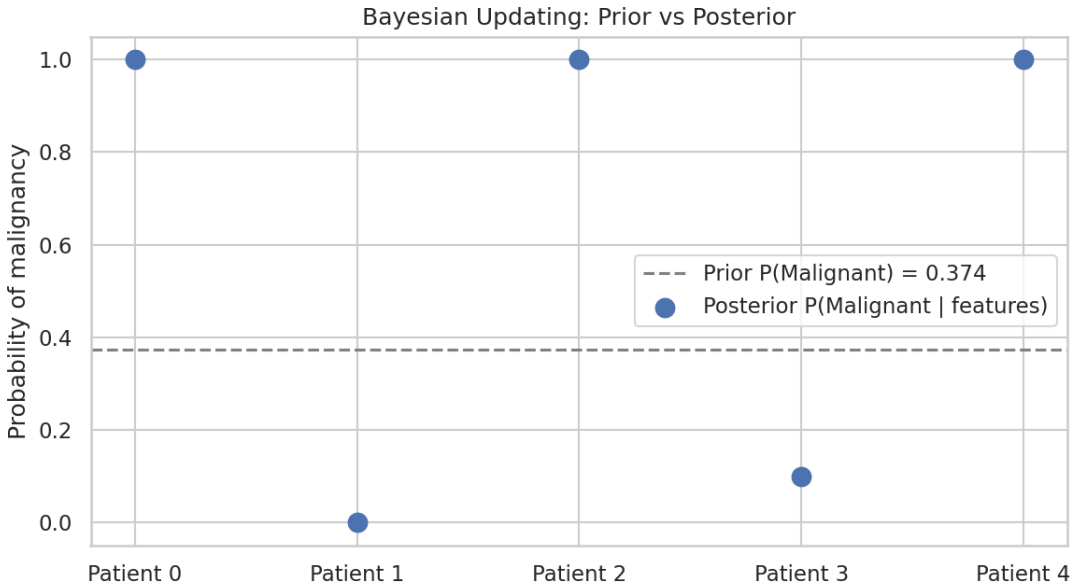
One Normal(μ, σ²) per feature and class; features independent

=

POSTERIOR

$$P(\text{class} \mid \text{features})$$

Prior × likelihood, normalised; the larger posterior is the prediction



How to read the plot

Dashed line: prior P(malignant) = 0.374. Dots: posterior after seeing a patient's 30 measurements. Four of five patients jump to about 0 or 1; Patient 3 (benign) lands at 0.098. Such extreme posteriors suggest the independence assumption is over-trusted.

Experimental pipeline

1 Load

569 x 30 table from scikit-learn (WDBC). Target: 0 = malignant, 1 = benign. No missing values.

2 Explore

Class balance, histograms with KDE (Gaussian check), boxplots by class, correlation heatmap.

3 Split

Stratified 80/20 split: 455 train, 114 test, random_state = 42, class ratio preserved.

4 Scale

StandardScaler fitted on training data only, then applied to the test set to avoid leakage.

5 Train

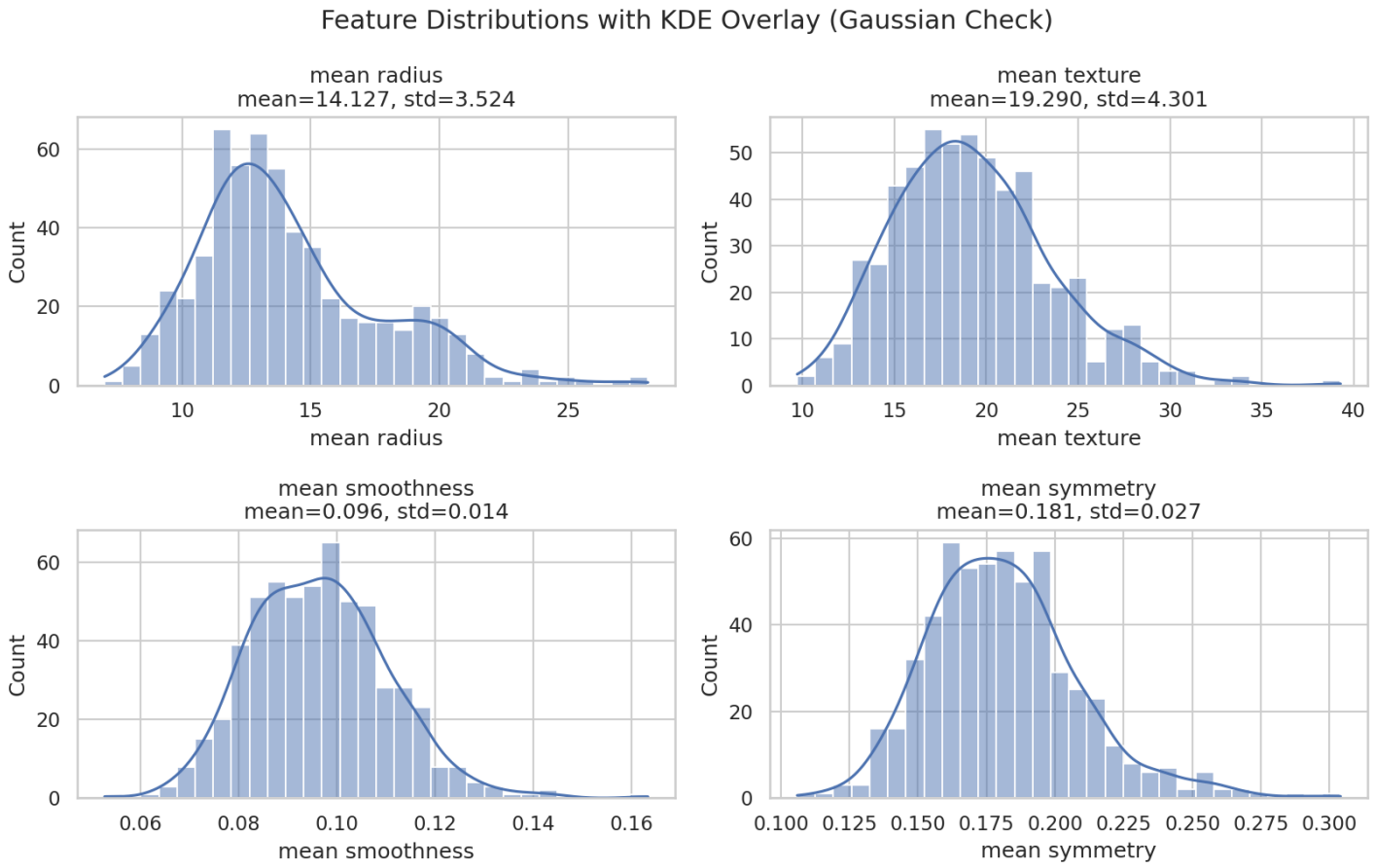
Logistic Regression (max_iter = 5000) and Gaussian Naive Bayes on the same scaled data.

6 Evaluate

Confusion matrix, precision / recall / F1, ROC-AUC, 5-fold CV with scaling inside each fold.

Tools: Python, pandas, NumPy, scikit-learn, matplotlib, seaborn. Run in Google Colab.

Exploratory analysis: is the data Gaussian?



What the histograms say

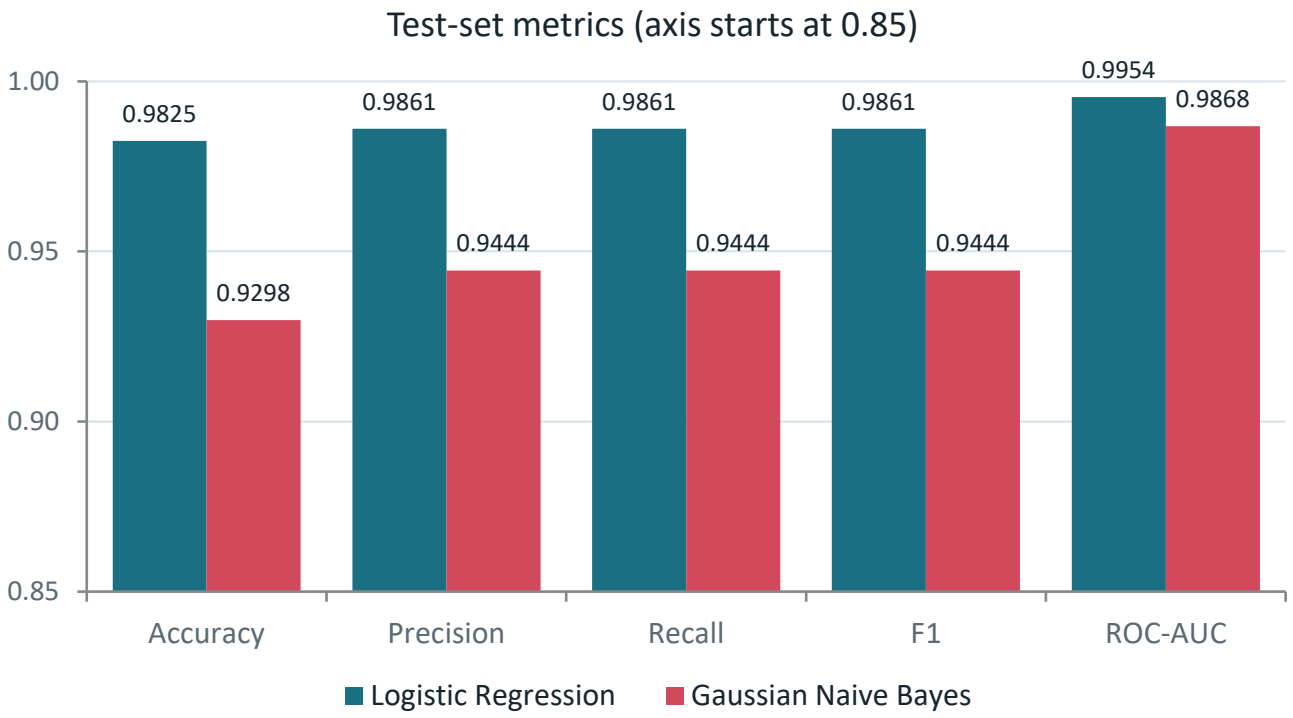
Roughly Gaussian: mean texture, smoothness and symmetry are bell-shaped with a mild right tail.

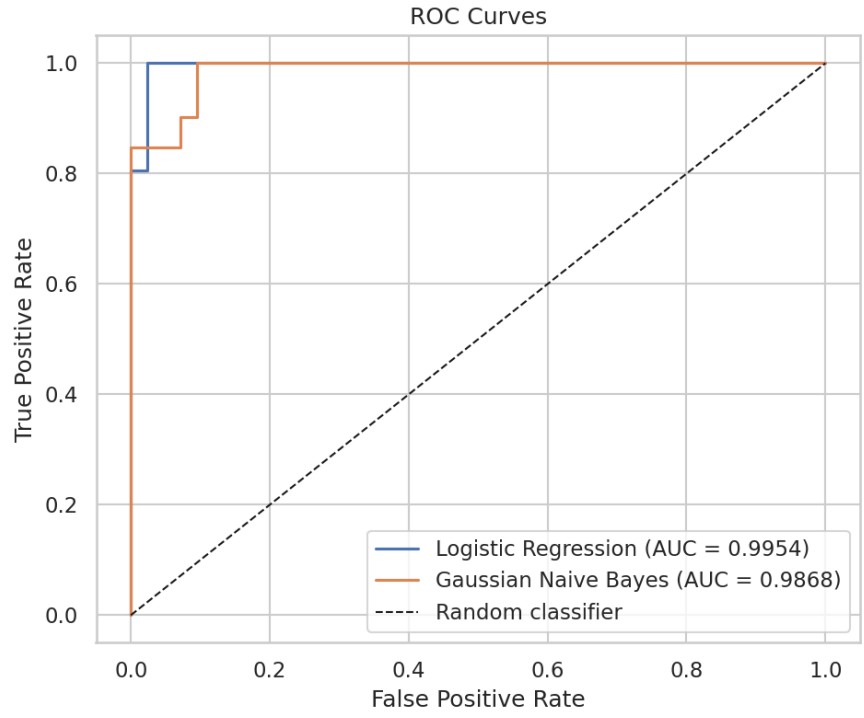
Not Gaussian: mean radius is right-skewed, because benign and malignant tumours overlap in one histogram.

Class matters: malignant tumours have larger radius and perimeter and more concave points (boxplots).

Redundant features: radius, perimeter and area correlate at r of about 0.99.

Logistic regression beats Naive Bayes



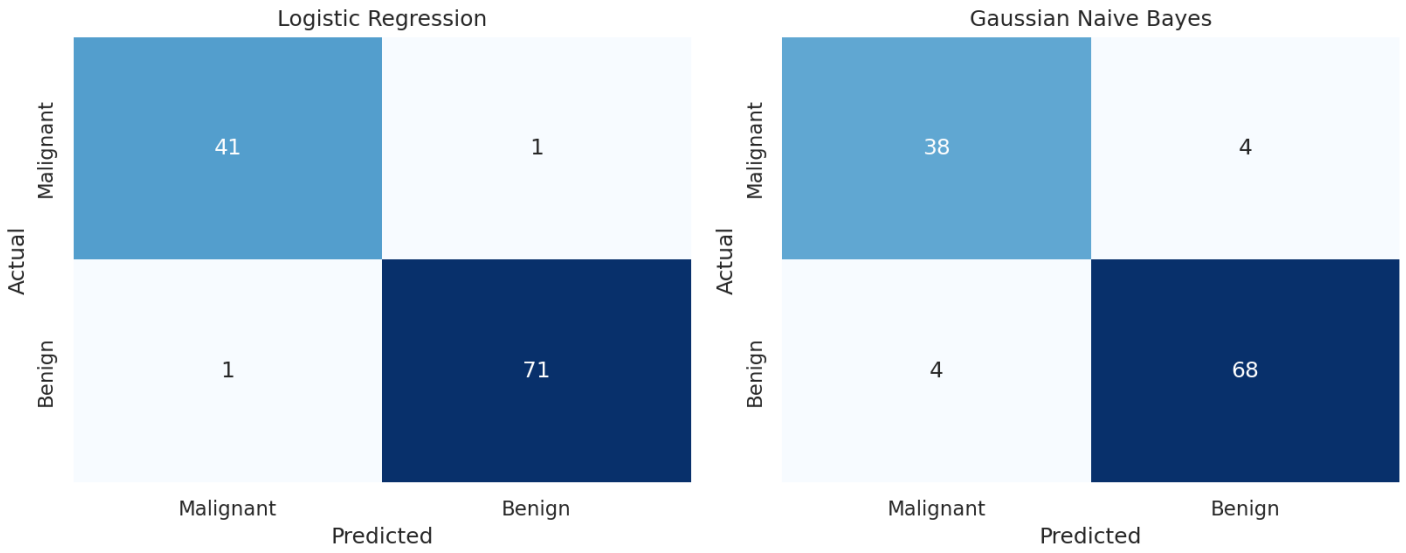


98.25% vs 92.98%
test accuracy (LR vs GNB)

0.9954 vs 0.9868
ROC-AUC (LR vs GNB)

0.9952 vs 0.9855
5-fold CV AUC (std 0.004 vs 0.010)

Error analysis: what each model missed



In screening, a missed malignancy costs far more than a false alarm.

1 of 42

malignant tumours missed by Logistic Regression (97.6% caught)

4 of 42

malignant tumours missed by Gaussian Naive Bayes (90.5% caught)

Conditional probability, by hand

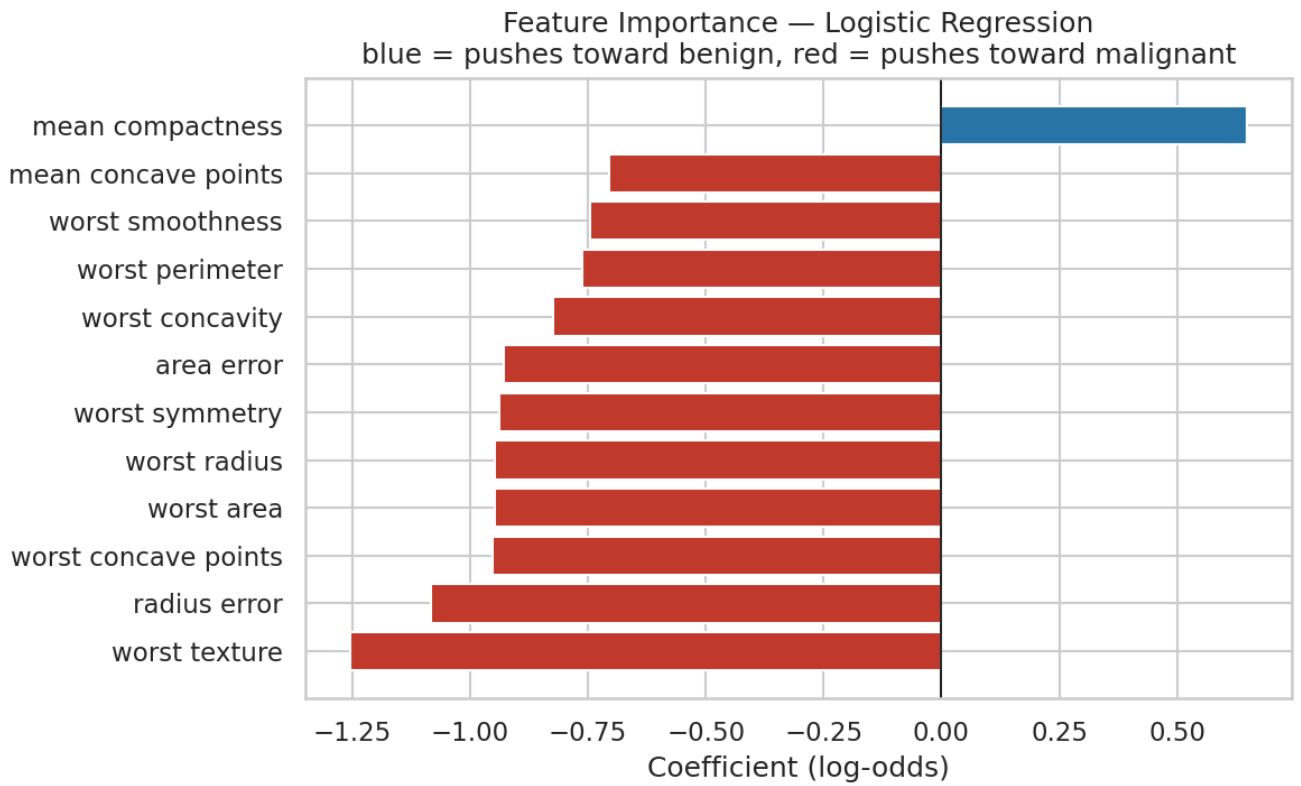
P(Benign | Predicted Benign)
= 71/72 = 0.9861
precision

P(Predicted Benign | Benign)
= 71/72 = 0.9861
recall (sensitivity)

P(Predicted Malignant | Malignant)
= 41/42 = 0.9762
specificity

Computed from the Logistic Regression confusion matrix; matches scikit-learn exactly.

What the model learned



Standardised coefficients (log-odds). Red pushes toward malignant, blue toward benign.

Reading a coefficient as an odds ratio

Feature	OR (benign)	Malignancy odds
worst texture	0.285	**x 3.51**
radius error	0.339	**x 2.95**
worst concave pts	0.385	**x 2.60**

Per +1 standard deviation, other features held fixed: a one-SD rise in worst texture multiplies the odds of malignancy by about 3.5.

Careful: mean compactness has a positive coefficient, likely a multicollinearity artefact (r = 0.88 with mean concavity). Read single coefficients with care.

Findings and limitations

What we found

- Logistic Regression wins on every metric: 98.25% vs 92.98% accuracy, and 0.9952 vs 0.9855 AUC in 5-fold CV.
- It missed 1 of 42 malignant tumours; Naive Bayes missed 4.
- Texture, concave-point and size features push predictions most strongly toward malignant.

Limitations

- One test set of 114 patients: 1 vs 4 misses is suggestive, not statistically tested.
- Correlated features (r about 0.99) break the independence assumption, a likely reason Naive Bayes trails.
- One well-known dataset, no external validation: not clinical evidence.
- Default hyper-parameters and a fixed 0.5 threshold.

Remaining work and timeline

1. Test whether the 98.3% vs 93.0% accuracy gap is real: McNemar's test on paired predictions.
2. Tune regularisation strength and the decision threshold to favour malignant recall; add a calibration plot.
3. Add a third comparison model (e.g. Random Forest) and a SHAP view of feature effects.
4. Write the IEEE-format report (8 to 10 pages): literature review, formulation, results, discussion.
5. Publish code and README on GitHub; submit the paper link and the final report.

TIMELINE

Project review

Problem, methods, initial results

20 Oct

Paper link submitted for plagiarism check

29 Oct

Final submission on VTOP